

Gynocamp Patient Registration Application

Developer Guide - what the app needs to do, in plain language

This document describes what the Gynocamp Patient Registration Application

Platforms	Android app + Web admin panel
User roles	3 — Super Admin, Data Taker, Data Analyst

Overview

Gynocamp runs periodic gynaecological health camps for women across Nepal. Camps are often held where there is little or no internet or electricity, so field staff need a way to record patient data reliably on-site and have it sync up later without anyone thinking about it.

This system is two connected pieces: an Android app that camp staff use to register patients and record clinical data in the field, and a web admin panel that staff back at the office use to manage camps, configure forms, and pull reports. Both talk to the same central database once data syncs.

User Roles

Every login belongs to one of three roles. Each role only sees and does what its job needs.

Role	What they do
Super Admin	Full control of the system — manages camps and staff, configures the clinical forms and medicine lists, and can see data across all camps.
Data Taker	Field role. Registers patients and enters clinical data during a camp, on a device that may be offline.
Data Analyst	Reviews camp data and generates reports once it has synced in.

Features

Grouped by what part of the system they belong to.

Patient & Data Entry

- **Patient registration:** Every patient gets a unique Patient ID, generated automatically at registration. No manual numbering.
- **Scan & auto-fill:** Staff can scan a completed paper form and the app fills in the digital fields for them. Manual typing is always available as a fallback.
- **Clinical data validation:** Numeric clinical fields — blood pressure, pulse, oxygen saturation — are checked against sensible ranges before saving, to catch typos at entry time.
- **Duplicate detector:** Before saving, the app cross-checks phone number, age, name, ward, and father's/husband's name against existing records and flags likely duplicates.

Offline & Sync

- **Offline-first storage:** The mobile app stores everything locally on the device and works fully without a connection — this is the normal case at camps, not an edge case.
- **Auto-sync:** As soon as the device reconnects, pending data syncs to the central server automatically. No manual upload step for field staff.

Security & Access

- **Role-based access:** Login and permissions are scoped to the three roles above — a Data Taker can't reach admin settings, for example.
- **Secure device access:** Only devices registered to the client can log in. A new device needs OTP verification and admin approval before it's activated.
- **App lock:** PIN or biometric lock on top of login, so a device left unattended at a camp doesn't expose patient data.
- **Activity log:** Every login, entry, and edit is recorded with the user's name and a timestamp — a full audit trail.

Admin & Customization

- **Camp management:** Create and manage camps, assign staff to each one, and see a staff schedule and camp calendar across all of them.
- **Form & medicine-list customization:** The Super Admin can update the clinical intake/examination form fields and medicine lists straight from the admin panel — no new app release needed. Digital forms are based on the client's existing paper form.

Reporting & Usability

- **One-tap camp reports:** Generate a full camp summary in one tap, exportable as PDF or Excel.
- **Bilingual interface:** Full Nepali and English UI throughout the app and admin panel.
- **Consistent, simple design:** Same design care on the mobile app and the admin panel, so non-technical field staff can use it without training.

Technology

Piece	What it's for
Android app	Used by Data Takers in the field. Offline-first, syncs when connected. Published on the Google Play Store.
Web admin panel	Browser-based dashboard for Super Admins to manage camps, staff, forms, and reports.
Central database	Secure, cloud-hosted store that becomes the single source of truth once field data syncs in.

An iOS/Apple App Store version isn't part of this build — it would be a separate add-on, scoped and quoted separately if the client needs it later.

Glossary

Term	Meaning
Patient ID	The unique identifier auto-generated for every registered patient.
OTP	One-Time Password — a short code used to verify a new device before it's allowed to log in.
Sync	Sending data stored offline on a device up to the central server once there is a connection.
Super Admin	The role with full system and configuration control.

Data Taker	The field role that registers patients and records clinical data at camps.
Data Analyst	The role that reviews synced data and produces reports.

Gynocamp Patient Registration System — developer feature reference. Keep this in sync as features are added or changed.